

RAG Automation API

A FastAPI Backend for Document Ingestion and Conversational
RAG

Technical Documentation

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Abstract

RAG Automation API is a modular FastAPI backend for document ingestion and retrieval-augmented generation. It accepts PDF and TXT documents, extracts and chunks their text using a selectable strategy, generates local BGE embeddings, and stores vectors in Qdrant. The conversational API retrieves relevant context without using `RetrievalQAChain`, preserves multi-turn chat history in Redis, and uses an LLM to detect and persist interview bookings in PostgreSQL.

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1 Overview

The project provides two REST APIs:

- **Document ingestion:** upload a `.pdf` or `.txt` file, extract text, select a chunking strategy, generate embeddings, store vectors, and save document metadata.
- **Conversational RAG:** ask document-grounded questions, retain Redis-backed conversation history, resolve follow-up questions, and store LLM-detected interview bookings.

The implementation does not use FAISS, Chroma, a user interface, or `RetrievalQAChain`. Retrieval and prompt construction are implemented directly in the application services.

1.1 Technology stack

Component	Technology	Purpose
API framework	FastAPI	Async REST endpoints
LLM	Gemini 2.5 Flash	Answers and booking extraction
Embeddings	BAAI/bge-base-en-v1.5	Local 768-dimensional vectors
Vector database	Qdrant	Similarity search
Session memory	Redis	Multi-turn chat history
Relational DB	PostgreSQL	Metadata and bookings
ORM and migrations	SQLAlchemy and Alembic	Persistence management
PDF parser	PyMuPDF	PDF text extraction
Chunking	Recursive and semantic	Structure and topic boundaries

2 Architecture

2.1 Ingestion flow

1. The client uploads a PDF or TXT document and selects `recursive` or `semantic` chunking.
2. FastAPI validates the extension and extracts text.
3. The chunker splits extracted text with the selected strategy.
4. The local BGE sentence-transformers model generates a vector for each chunk on CPU.
5. Qdrant stores vectors with document ID, filename, strategy, chunk index, chunk count, and chunk text.
6. PostgreSQL stores document metadata.

2.2 Conversational RAG flow

1. The client sends a `session_id` and query.
2. Redis loads the recent conversation history for that session.
3. Gemini checks the current query and history for interview-booking intent and the required fields.
4. If a complete booking is found, PostgreSQL stores it and the API returns a confirmation.
5. Otherwise, the local BGE model embeds the query with its retrieval instruction prefix.
6. Qdrant compares the query vector with stored chunk vectors using cosine similarity and returns the five most similar chunks.
7. Retrieved text is normalized before prompt construction so legacy PDF line wrapping does not appear in generated answers.
8. The application builds a grounded prompt manually and Gemini generates the answer.
9. Redis stores the user message and assistant response.

3 Environment setup

3.1 Prerequisites

- Python 3.11 or higher
- Docker and Docker Compose
- A Gemini API key from [Google AI Studio](#)

3.2 Installation

```
git clone https://github.com/ishaa-bht/palm-mind-rag.git
cd palm-mind-rag

python -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

The local BGE model is downloaded and cached automatically on first use.

Create a `.env` file in the project root:

```
GEMINI_API_KEY=your_gemini_api_key_here
DATABASE_URL=postgresql+asyncpg://palmuser:palmpass@localhost:5433/palmdb
QDRANT_HOST=localhost
QDRANT_PORT=6333
QDRANT_COLLECTION_NAME=documents_bge_base_en_v1_5
REDIS_URL=redis://localhost:6379
APP_ENV=development
```

```
EMBEDDING_MODEL_NAME=BAAI/bge-base-en-v1.5
EMBEDDING_DEVICE=cpu
EMBEDDING_BATCH_SIZE=32
EMBEDDING_DIM=768
```

Start infrastructure, apply migrations, and run the API:

```
docker compose up -d
alembic upgrade head
uvicorn app.main:app --reload
```

Swagger UI is available at <http://localhost:8000/docs>.

3.3 Gemini access errors

If chat returns 503 **Service Unavailable** with a Gemini access-denied message, verify `GEMINI_API_KEY` and the Google AI project status, then restart the API. This indicates an upstream project-access problem.

4 Module reference

Module	Responsibility
<code>app/main.py</code>	Creates the FastAPI app, registers routers, and initializes the Qdrant collection during startup.
<code>app/api/ingest.py</code>	Implements PDF and TXT ingestion, extraction, chunking, embedding, vector upsert, and metadata persistence.
<code>app/api/chat.py</code>	Implements conversational retrieval, Redis memory, LLM answers, and booking persistence.
<code>app/services/chunker.py</code>	Provides recursive structure-aware chunks and locally embedded semantic topic chunks.
<code>app/services/embedder.py</code>	Generates document and query embeddings locally with BAAI/bge-base-en-v1.5 on CPU.
<code>app/services/vector_store.py</code>	Creates the Qdrant collection, upserts chunks with retrieval metadata, normalizes retrieved PDF text, and returns the top five matches using cosine similarity.
<code>app/services/memory.py</code>	Stores up to 20 Redis messages per session with a one-hour inactivity TTL.
<code>app/services/llm.py</code>	Builds the custom grounded prompt and generates answers without <code>RetrievalQAChain</code> .
<code>app/services/booking.py</code>	Extracts name, email, date, and time from conversational booking intent.
<code>app/models/db_models.py</code>	Defines document metadata and booking SQLAlchemy models.

4.1 Chunking strategies

Strategy	Configuration	Recommended use
<code>recursive</code>	Paragraph, sentence, and word fallbacks; maximum 300 words with 45-word overlap	General-purpose ingestion
<code>semantic</code>	Batched local BGE sentence embeddings; cosine boundary threshold of 0.65; recursive fallback for oversized chunks	Multi-topic reports and research

4.2 Vector retrieval

Qdrant stores one 768-dimensional local BGE embedding for each document chunk. During chat, the application embeds the user's query and performs a **cosine-similarity search**. Qdrant ranks stored chunk vectors by their directional similarity to the query vector, and the API sends the top five matching chunks to Gemini as grounded context. Each Qdrant payload stores `doc_id`, `filename`, `chunk_index`, `chunk_count`, `strategy`, and `text`. Retrieved text is normalized before it reaches the LLM prompt to remove PDF line-wrap artifacts.

Parameter	Value
Vector database	Qdrant
Distance metric	Cosine similarity
Embedding dimensions	768
Retrieved chunks per query	5

4.3 Database schema

Table	Stored fields
<code>document_metadata</code>	<code>id</code> , <code>filename</code> , <code>file_type</code> , <code>chunking_strategy</code> , <code>chunk_count</code> , <code>uploaded_at</code>
<code>bookings</code>	<code>id</code> , <code>name</code> , <code>email</code> , <code>date</code> , <code>time</code> , <code>session_id</code> , <code>created_at</code>

5 API reference

5.1 POST /api/v1/ingest

Uploads and processes a document using multipart form data.

Field	Type	Description
file	File	Required PDF or TXT file
strategy	String	Required: recursive or semantic

```
curl -sS -X POST http://localhost:8000/api/v1/ingest \
  -F "file=@test_doc.txt" \
  -F "strategy=recursive" \
  | python -m json.tool
```

5.2 POST /api/v1/chat

Answers questions using retrieved document context and Redis session history.

```
curl -sS -X POST http://localhost:8000/api/v1/chat \
  -H "Content-Type: application/json" \
  -d '{"session_id":"user-001","query":"What skills does John have?"}' \
  | python -c 'import json, sys; print(json.load(sys.stdin)["answer"])
```

5.3 Health endpoints

Endpoint	Response
GET /	API status and message
GET /health	{"status":"healthy"}

6 Testing

6.1 Health check

```
curl -sS http://localhost:8000/health | python -m json.tool
```

6.2 Ingestion with both strategies

```
curl -sS -X POST http://localhost:8000/api/v1/ingest \
  -F "file=@test_doc.txt" -F "strategy=recursive" \
  | python -m json.tool

curl -sS -X POST http://localhost:8000/api/v1/ingest \
  -F "file=@test_doc.txt" -F "strategy=semantic" \
```



```
| python -m json.tool
```

6.3 Retrieval-grounded answer

```
curl -sS -X POST http://localhost:8000/api/v1/chat \
-H "Content-Type: application/json" \
-d '{"session_id":"test-001","query":"What skills does John have?"}' \
| python -c 'import json, sys; print(json.load(sys.stdin)["answer"])
```

6.4 Multi-turn memory

Use the same `session_id` for a follow-up question:

```
curl -sS -X POST http://localhost:8000/api/v1/chat \
-H "Content-Type: application/json" \
-d '{"session_id":"test-001","query":"Where did he work?"}' \
| python -c 'import json, sys; print(json.load(sys.stdin)["answer"])
```

6.5 Interview booking

```
curl -sS -X POST http://localhost:8000/api/v1/chat \
-H "Content-Type: application/json" \
-d '{"session_id":"booking-001","query":"Book an interview for Jane Doe, jane@example.com, on 2026-06-10 at 14:00."}' \
| python -c 'import json, sys; print(json.load(sys.stdin)["answer"])
```

Inspect saved records from the PostgreSQL container:

```
docker compose exec postgres \
psql -U palmuser -d palmdb \
-c "SELECT name, email, date, time, session_id, created_at FROM bookings ORDER BY created_at DESC LIMIT 5;"
```

6.6 Sample PDF demonstration

The repository includes `sample.pdf` for a copy-pasteable demonstration of document ingestion and grounded question answering. Ingest it with:

```
curl -sS -X POST http://localhost:8000/api/v1/ingest \
-F "file=@sample.pdf" \
-F "strategy=recursive" \
| python -m json.tool
```

Then reuse `sample-pdf-session` while asking:

1. What is the title of the research paper?

2. What percentage of students started using mobile devices for English learning at university?
3. How did Holec define learner autonomy?
4. Why did students prefer using mobile devices for learning English?
5. What evidence suggests that mobile devices promoted autonomous learning?

For example:

```
curl -sS -X POST http://localhost:8000/api/v1/chat \
  -H "Content-Type: application/json" \
  -d '{"session_id":"sample-pdf-session","query":"How did Holec define learner
    autonomy?"}' \
  | python -c 'import json, sys; print(json.load(sys.stdin)["answer"])'
```

7 Design decisions

- **Custom RAG pipeline:** retrieval, context formatting, and prompting are explicit application steps instead of a framework chain.
- **Qdrant vector store:** vectors and payload metadata are persisted in a dedicated similarity-search database. Cosine similarity is used to retrieve the five chunks most relevant to each query.
- **Redis session memory:** short-lived chat history supports follow-up questions while keeping the API stateless.
- **PostgreSQL persistence:** structured document and booking records are managed with SQLAlchemy and Alembic.
- **Selectable chunking:** callers can choose efficient recursive boundary-aware chunks or topic-aware semantic detection based on retrieval-quality needs.
- **Local semantic analysis:** semantic mode embeds sentences in CPU batches of 32 texts using BAAI/bge-base-en-v1.5. This removes embedding API cost and rate limits while retaining topic-aware boundaries.

8 Conclusion

RAG Automation API meets the core backend requirements for document ingestion, custom retrieval-grounded conversation, Redis-backed multi-turn memory, and LLM-assisted interview booking. Its modular service boundaries allow each integration to be tested and improved independently while keeping the public API compact.